

ALISSA ANN JOSY

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EDUCATION

University of South Florida

Tampa, FL

Bachelor of Science in Computer Science, GPA: 3.78 / 4.0

May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Artificial Intelligence, Large Language Models, Software Engineering, Database Design, Computer Architecture.

EXPERIENCE

Software Engineering Intern

May 2026 – Aug 2026

Publix Technology

- Engineered a production Kafka-based C#/NET pipeline and APIs replacing a slow, unreliable legacy workflow, reducing invoice cost-resolution time to under 2 seconds for 10,000+ daily invoices.
- Architected a 3-tier fault-tolerant fallback across distributed cache, SQL Server, and external APIs, maintaining uninterrupted billing during single-source outages and a staged multi-warehouse rollout.
- Delivered 22 production requirements across 7 user stories with 84 xUnit/Moq tests, collaborating in an Agile engineering team and enforcing correctness through spec-driven development and CI/CD.

Software/AI Research Assistant

Aug 2025 – Jan 2026

RARE Lab at USF

- Shipped a production Python/Gemini backend powering a live robotic recommendation experience, serving 200+ real user requests with personalized multimodal AI generated outputs.
- Reduced LLM API failure from 20% to under 3% by diagnosing race conditions in concurrent request pipelines; implemented a real-time WebSocket communication between the AI backend and tablet interface with sub-second latency.

Student Assistant

June 2024 - Present

Undergraduate Studies at USF

- Automated advisor assignment using Python and PostgreSQL, reducing processing time by 20% and support tickets by 40%.
- Maintained an ETL data pipeline handling 300+ weekly requests, resolving data-quality and parsing edge cases to improve workflow reliability.

PROJECTS

Backtest Auditor – Nebius x NVIDIA AI Hackathon | *Python, NVIDIA Nemotron, Sandboxes, Pandas, NumPy, Streamlit*

- Built an autonomous AI agent that detects lookahead bias, data leakage, and overfitting in trading-strategy code using deterministic adversarial tests executed in isolated Nebius cloud sandboxes.
- Designed a write-run-verify repair loop where Nemotron generates code fixes, evaluates them across parallel sandbox branches, and accepts only fixes that pass behavioral tests; added structured-output guardrails, evidence validation, and 100+ automated tests.

Distributed Key-Value Store | *Go, TCP Sockets, RESP Protocol, Consistent Hashing, Docker*

- Implemented and deployed a Redis-compatible distributed key-value store from scratch, using TCP/RESP communication and consistent hashing to route 10,000+ keys across a live 3-node cluster.
- Designed consistent hashing with 150 virtual nodes, reducing key reshuffling from 75% to 25% during topology changes; added AOF persistence and asynchronous replication and benchmarked 10,101 SET and 21,707 GET ops/sec.

CollabPad – Real-Time Collaborative Editor | *FastAPI, WebSockets, PostgreSQL, Next.js, Monaco Editor*

- Engineered a real-time collaborative editor supporting 25+ simultaneous users, implementing WebSocket broadcasting, shared document state and concurrent conflict resolution with sub-50ms end-to-end latency.
- Optimized p99 message latency by 73% (45ms to 12ms) through multicore processing, connection pooling, and PostgreSQL indexing; load tested at 3,500+ messages/minute.

ACTIVITIES & EXTRACURRICULAR

President

April 2026 - Present

Bridges International at USF

- Lead a 6-person cross-functional team serving 200+ students; grew event attendance from 60 to 200 and created Python automation reducing weekly manual work by 90%.

Instructor

August 2025

Girls Who Code at USF

- Taught object-oriented programming and algorithms to 40 students, helping students apply programming concepts through independent software projects.

TECHNICAL SKILLS

LANGUAGES: Python, Java, JavaScript / TypeScript, Go, C#/NET, C/C++, SQL

WEB & BACKEND: React, Next.js, Node.js, Express, FastAPI, REST APIs, WebSockets, Kafka, PostgreSQL, MySQL, MongoDB, SQL Server, distributed systems.

AI/LLM: Google Gemini API, Groq API, Ollama, LLM evaluation, multimodal AI, prompt engineering, multimodal AI, NLP, AI output validation.

TOOLS: Git/GitHub, Docker, CI/CD, Azure DevOps, Linux, Google Cloud Platform (GCP), Jira, Agile/Scrum, test-driven development, unit/integration testing.